

Policy on Imaging Quality Assurance and Safety Programme

«Hospital Name»

Document control

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Prepared by	Imaging In-Charge Name: _____ Signature: _____		
Reviewed by	Quality Coordinator Name: _____ Signature: _____		
Approved by	Medical Superintendent Name: _____ Signature: _____		

Statement of intent

«Hospital Name» implements a comprehensive quality assurance and safety programme for imaging services, meeting statutory AERB requirements, screening patients before radiation or MRI exposure, using and testing radiation-safety devices, and training imaging and ancillary staff in safety practices.

1. Purpose

This policy explains how «Hospital Name» maintains an imaging quality assurance and safety programme: equipment and protocol quality assurance, appropriateness screening of investigation orders, peer review of imaging results, clinico-radiological meetings, CAPA on QA deviations, the radiation-safety programme and ALARA principle, pre-imaging safety screening, radiation-safety device use and testing, staff training, and imaging signage.

Informed consent for contrast, sedation or interventional procedures is covered in the hospital's other policies. AERB clearance and PC-PNDT registration for imaging services are covered in the hospital's other policies.

2. Scope

This policy applies to the Imaging In-Charge, radiologists, radiographers, MRI/CT technologists, ancillary imaging staff, and all patients and attendants entering imaging areas.

3. Policy standards

«Hospital Name» implements a comprehensive quality assurance programme addressing imaging equipment, protocols, surveillance and safety, meeting statutory AERB requirements, and including CAPA for deviations.

A radiation-safety programme is implemented following AERB guidelines and the ALARA principle. Patients and attendants are appropriately screened for safety and risk before imaging. Radiation-safety and monitoring devices are used and periodically tested. Imaging and ancillary personnel are trained in safety practices, and imaging signage is prominently displayed.

4. Non-negotiable rules

- Do not proceed with radiation or MRI exposure when required safety/risk screening has not been completed and documented for the patient, or for any attendant entering the imaging area.
- Do not proceed with a radiation examination when a required radiation-safety or monitoring device has failed its periodic test — for example a cracked lead apron — until it is repaired or replaced.
- Do not proceed with a radiation examination when required TLD badges are not available for staff working with radiation sources.
- Do not order routine, unjustified radiation exposure — for example a routine daily chest radiograph for all ICU patients without clinical indication, especially in paediatric or neonatal ICUs.
- Do not display imaging signage inconsistently with regulatory requirements or omit it from procedure rooms and operation theatres where imaging equipment is used.

5. What we do

5.1 Implement the imaging quality assurance programme

A comprehensive quality assurance programme for imaging services is implemented, addressing equipment, protocols, surveillance and safety, and meeting statutory AERB requirements. Equipment tests are performed as applicable to the modalities present. Quality assurance for nuclear medicine and radiotherapy, where those services exist, follows AERB guidelines. Imaging protocols are reviewed to ensure optimum image quality with minimum possible dosage.

5.2 Screen investigation orders for appropriateness

A system screens investigation orders before imaging or interventional procedures to ensure they are appropriate for the clinical indication, in terms of current best-practice guidelines and patient safety. If a request is not appropriate, alternate investigations are offered in consultation with the treating doctor. A record is maintained of instances where the requested test was modified after consultation.

5.3 Conduct peer review of imaging results

A peer review system reviews imaging results of CT and MRI in a structured manner. Sample size and periodicity for each modality are defined by the organisation, at a minimum one percent. Discrepancies are graded on severity and impact on patient management, and CAPA to minimise recurrence is documented.

5.4 Hold clinico-radiological meetings

Clinico-radiological meetings are conducted at pre-defined intervals to correlate imaging results — at a minimum CT and MRI — with referring clinicians, and are used to improve the quality of imaging results.

5.5 Document CAPA for QA deviations

When deviations from the laid-down quality assurance programme are noted, the organisation institutes corrective and preventive action as appropriate, and this is documented as part of the QA programme.

5.6 Implement the radiation-safety programme

The radiation-safety programme follows AERB guidelines and implements the As Low As Reasonably Achievable (ALARA) principle in all investigations involving radiation, including screening of patients at high risk for radiation. Routine unjustified radiation is avoided. CT protocols use the lowest exposure parameters that maintain image quality appropriate for the clinical indication. The programme is aligned with the organisation's safety programme.

5.7 Screen patients for safety and risk before imaging

Patients in the child-bearing age group requiring radiation exposure are screened for pregnancy before the examination. Patients undergoing MRI are screened for any magnetic substance. Screening is documented and applies to attendants accompanying a patient or child into the imaging area. Informed consent is taken for contrast injection, moderate-deep sedation, interventional procedures, and higher-risk findings on risk screening.

5.8 Use radiation-safety and monitoring devices

Imaging personnel and patients use appropriate radiation-safety and monitoring devices where applicable. Shielding of body parts of staff, patients and attendants uses appropriate aprons and shields, in numbers adequate for all workers. Staff directly working with radiation sources possess and use thermo-luminescent dosimeter (TLD) badges.

5.9 Test radiation-safety devices periodically

Radiation-safety and monitoring devices are periodically tested and results documented. Protective devices such as lead aprons are exposed to X-ray, fluoroscopy or CT scout view to verify cracks and damage; images of the checks are stored. CAPA is taken and documented where a device test result warrants it. TLD badges are replaced by fresh badges from accredited laboratories at the frequency recommended by AERB.

5.10 Train imaging and ancillary personnel

Imaging personnel are trained in imaging safety practices — such as MRI safety, tube kinking, fall prevention and patient handling — and radiation-safety measures. Ancillary staff posted in imaging, including nurses, helpers, stretcher-bearers, housekeeping and security as applicable, are also trained.

5.11 Display imaging signage

Imaging signage, including safety signage and signage required by regulatory authorities, is prominently displayed in all appropriate locations — including procedure rooms and operation theatres where imaging equipment is used.

6. Stop-work authority

Do not proceed with radiation or MRI exposure when required safety/risk screening has not been completed and documented for the patient (and for any attendant entering the imaging area).

As hospital safety practice (not an NABH objective-element requirement), do not proceed when pregnancy status is positive or uncertain for a radiation examination until the hospital radiologist or imaging in-charge has reviewed and documented the decision. Do not proceed with MRI when magnetic-substance screening is incomplete or positive for a contraindication.

Do not proceed when a required radiation-safety or monitoring device has failed its periodic test (for example a cracked lead apron) or when required TLD badges are not available for staff working with radiation sources.

Stop-work applies to the imaging exposure or MRI examination, not to emergency clinical stabilisation outside the imaging room.

The person who stops tells the Imaging In-Charge or the Radiation Safety Officer the same shift. Refusing to expose without screening or with failed safety devices is not a disciplinary matter.

7. Governance and responsibility

Role	Responsibility
Medical Superintendent	Accountable for the imaging quality assurance and safety programme; receives stop-work escalations.
Imaging In-Charge	Owns implementation of this policy; maintains the QA programme, appropriateness screening system, peer review, and clinico-radiological meeting schedule.
Radiation Safety Officer	Oversees the radiation-safety programme, ALARA implementation, device testing, and TLD badge management; receives stop-work escalations.

Radiologists/Radiographers/Technologists	Perform pre-imaging screening, use safety devices, and follow appropriateness and ALARA protocols.
Quality Coordinator	Audits this policy; holds training records, CAPA logs, and stop-work event logs.

8. Quality monitoring

Monitoring area	What is monitored
QA programme	Equipment test records; protocol review file; nuclear medicine/radiotherapy QA records where applicable.
Appropriateness screening	Modification log; consultation records with treating doctors.
Peer review	Discrepancy grading records; CAPA for recurring findings.
Clinico-radiological meetings	Meeting minutes; quality improvement actions arising.
CAPA on deviations	CAPA log with root cause, owner, and closure evidence.
Radiation-safety programme	ALARA protocol records; unjustified-exposure avoidance records.
Pre-imaging screening	Pregnancy and magnetic-substance screening records; consent records.
Safety device use and testing	Apron/shield inventory; TLD issue and wear records; periodic test images and results.
Training	Induction and periodic training records for imaging and ancillary staff.
Signage	Signage location checklist; periodic visibility check log.

This policy is reviewed annually, or sooner if NABH standards or applicable AERB regulations change.

9. Training and staff acknowledgement

All imaging and ancillary personnel shall be trained in imaging safety practices and radiation-safety measures, including the stop-work triggers in this policy, at induction and periodically.

I have read the Policy on Imaging Quality Assurance and Safety Programme of «Hospital Name». I will follow the processes described.

10. Distribution

This policy shall be available to all imaging and ancillary staff, the Radiation Safety Officer, the Imaging In-Charge, and the Quality Coordinator.

11. Abbreviations

Abbreviation	Meaning
AERB	Atomic Energy Regulatory Board
ALARA	As Low As Reasonably Achievable
CAPA	Corrective and Preventive Action
CT	Computed Tomography
MRI	Magnetic Resonance Imaging
NABH	National Accreditation Board for Hospitals and Healthcare Providers
TLD	Thermo-Luminescent Dosimeter

12. Traceability table

This table is an index. It is not how the policy is organised. An asterisk in the Level column means the objective element is starred and requires documented evidence.

Objective Element	Level	Traceability to this policy
AAC.9.a	Commitment*	Sections 3 and 5.1 address the imaging QA programme.
AAC.9.b	Achievement	Section 5.2 addresses appropriateness screening of investigation orders.
AAC.9.c	Achievement	Section 5.3 addresses peer review of imaging results.
AAC.9.d	Excellence	Section 5.4 addresses clinico-radiological meetings.
AAC.9.e	Commitment*	Section 5.5 addresses CAPA documentation for QA deviations.
AAC.9.f	Commitment*	Sections 3 and 5.6 address the radiation-safety programme and ALARA principle.
AAC.9.g	Commitment	Sections 3, 5.7 and 6 address pre-imaging safety screening and the related stop-work trigger.
AAC.9.h	Commitment	Section 5.8 addresses use of radiation-safety and monitoring devices.
AAC.9.i	Commitment*	Sections 5.9 and 6 address periodic testing of safety devices and the related stop-work trigger.
AAC.9.j	Commitment	Section 5.10 addresses training of imaging and ancillary personnel.
AAC.9.k	Commitment	Section 5.11 addresses imaging signage.

13. Required records/evidence checklist

QA programme — AAC.9.a (Commitment*)

- QA programme document, equipment test records, protocol review file.
- Nuclear medicine/radiotherapy QA records where applicable.

Appropriateness screening — AAC.9.b (Achievement)

- Appropriateness screening method or checklist.
- Modification log and consultation records with treating doctors.

Peer review — AAC.9.c (Achievement)

- Peer-review records with discrepancy grading and CAPA.
- Discrepancy meeting notes where held.

Clinico-radiological meetings — AAC.9.d (Excellence)

- Meeting schedule and minutes recording correlations with clinicians.
- Documented quality improvement actions.

CAPA on QA deviations — AAC.9.e (Commitment*)

- CAPA log for QA and radiation-safety deviations with root-cause notes, owners, due dates and closure evidence.

Radiation-safety programme — AAC.9.f (Commitment*)

- Radiation-safety programme document aligned with AERB and organisation safety.
- ALARA protocol examples and unjustified-exposure avoidance records.

Pre-imaging screening — AAC.9.g (Commitment)

- Documented pregnancy screening and MRI magnetic-substance screening.
- Consent records for contrast/sedation/interventional procedures.

Safety device use — AAC.9.h (Commitment)

- Apron and shield inventory adequate for all workers.
- TLD badge issue and wear records.

Safety device testing — AAC.9.i (Commitment*)

- Periodic lead-apron/shield test images and results.
- TLD replacement records per AERB-recommended frequency.
- CAPA/removal-from-use records for failed devices.

Staff training — AAC.9.j (Commitment)

- Induction and periodic training records for imaging and ancillary staff, with content and attendance lists.

Imaging signage — AAC.9.k (Commitment)

- Signage location checklist including procedure rooms and OTs with imaging equipment.

- Periodic signage visibility check log.

14. References

1. National Accreditation Board for Hospitals and Healthcare Providers. NABH Accreditation Standards for Hospitals, 6th Edition. AAC.9.
2. Atomic Energy Regulatory Board, Government of India. (2004).
<https://www.aerb.gov.in/images/PDF/RPR2004-with-corrigendum.pdf> Atomic Energy (Radiation Protection) Rules, 2004 — governing radiation licensing, protection, and safety compliance for radiation-emitting imaging services.

Disclaimer

This document is a template prepared for the guidance of «Hospital Name» and must be reviewed, adapted and formally approved by «Hospital Name» before use. Every entry marked _____ must be completed before the document is issued.

This policy references the Atomic Energy (Radiation Protection) Rules, 2004, as the statutory framework for radiation-safety duties to the extent «Hospital Name»'s imaging services fall within its scope. This policy is not a statement of, or substitute for, compliance with that or any other statute; «Hospital Name» remains separately responsible for meeting all applicable statutory and regulatory requirements, including AERB clearance and monitoring obligations.

The clinical and technical content reflects recognised national and international guidance current at the date of preparation. «Hospital Name» remains responsible for verifying that it is current and consistent with the edition of the accreditation standard against which it is being assessed.

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